New Approaches: Wound Dressings for Low Resource Settings

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May 2025

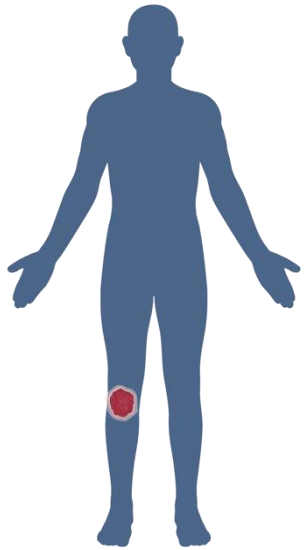
BE 183/283, spring 2025

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potassium	arsenic	potassium	oxygen	lanthanum	boron
19	33	19	8	57	5
K	As	K	O	La	B
39.098	74.922	39.098	15.9994	138.91	10.811

Field polymerizable hydrogels for battle-field wounds

KEY BATTLEFIELD CHALLENGES



Blast injuries leading to amputations

Polymicrobial infections (*S. aureus, E. coli, P. aeruginosa*)



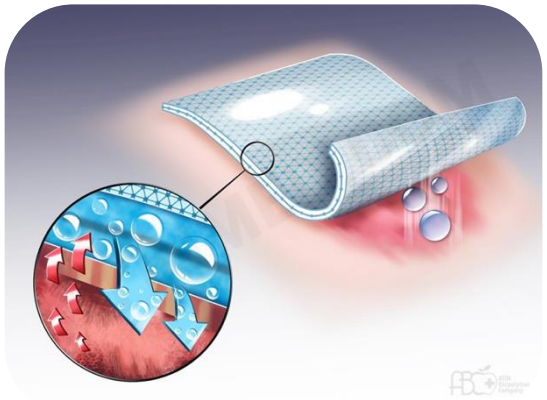
Bleeding and pain control

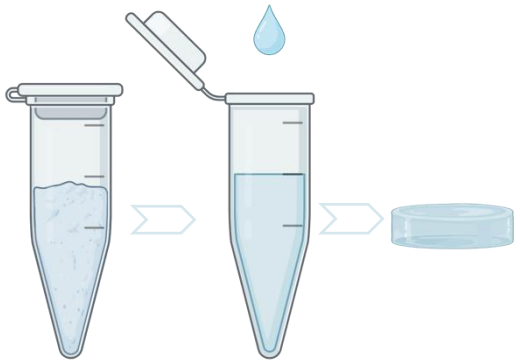
PROLONGED FIELD CARE



Need: light-weight, durable, addresses polymicrobial infections, pain & bleeding

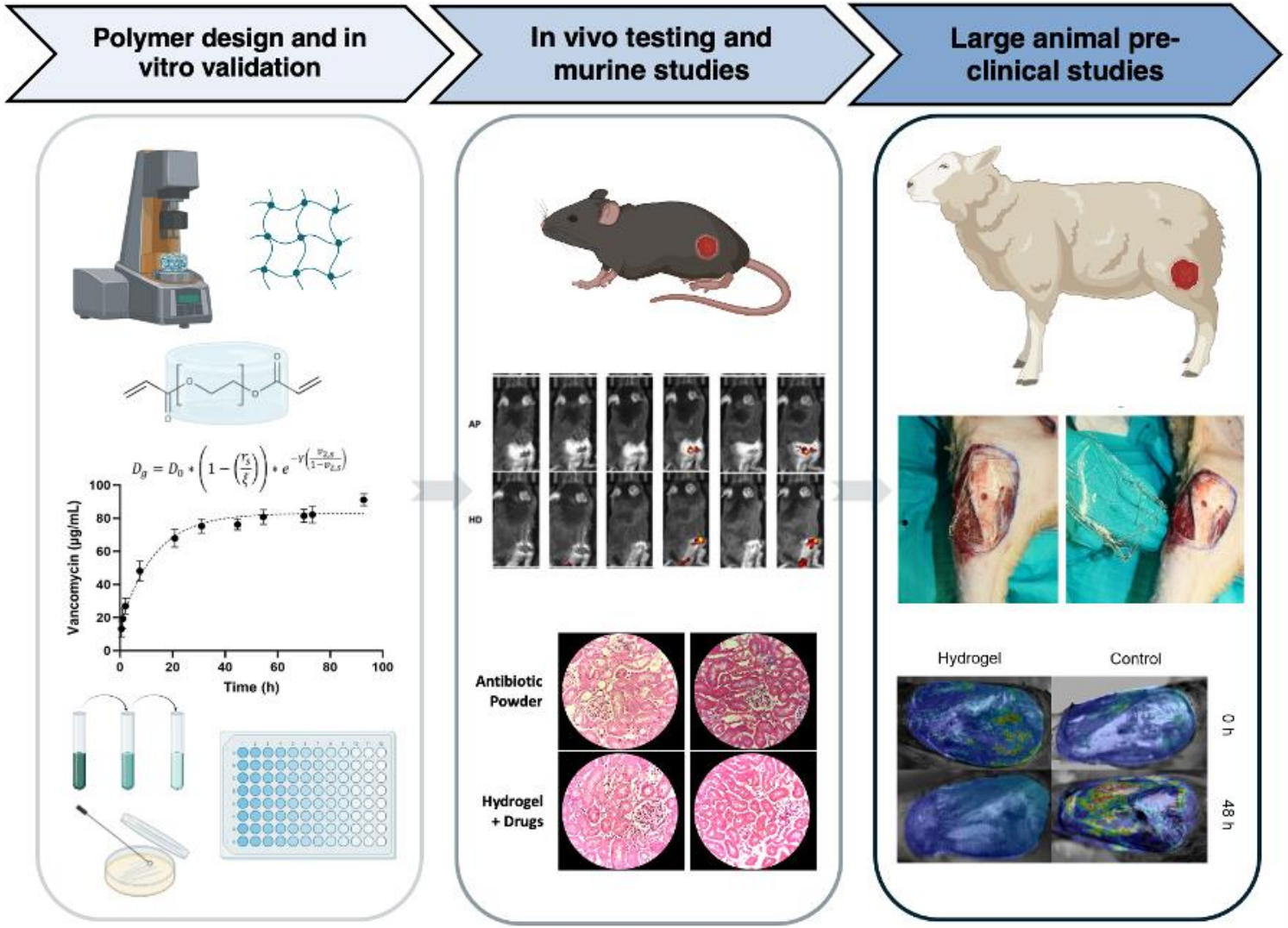
HYDROGEL-BASED SOLUTION





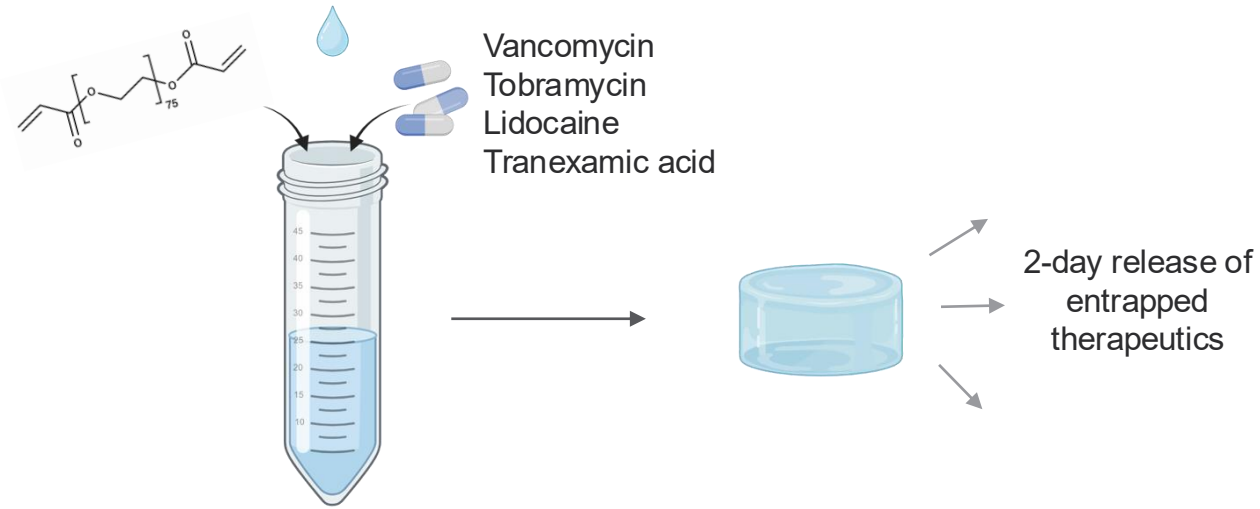
Powder → Liquid → Hydrogel (minutes)

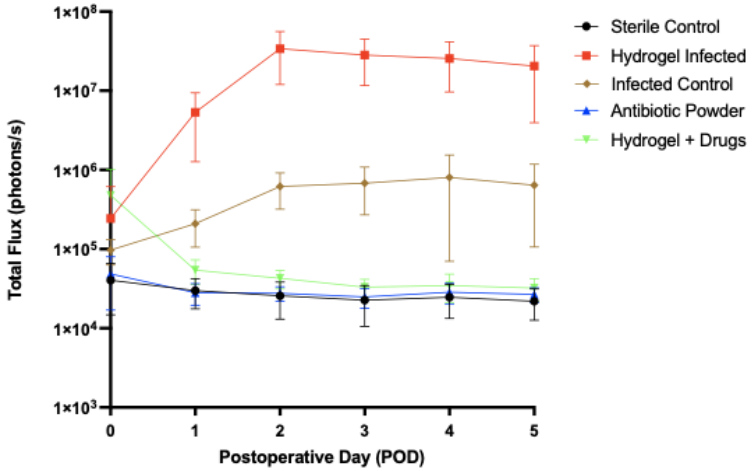
Field polymerizable hydrogels for battle-field wounds



Field polymerizable hydrogels for battle-field wounds

TRANSLATION FROM BENCHTOP TO IN VIVO MODELS

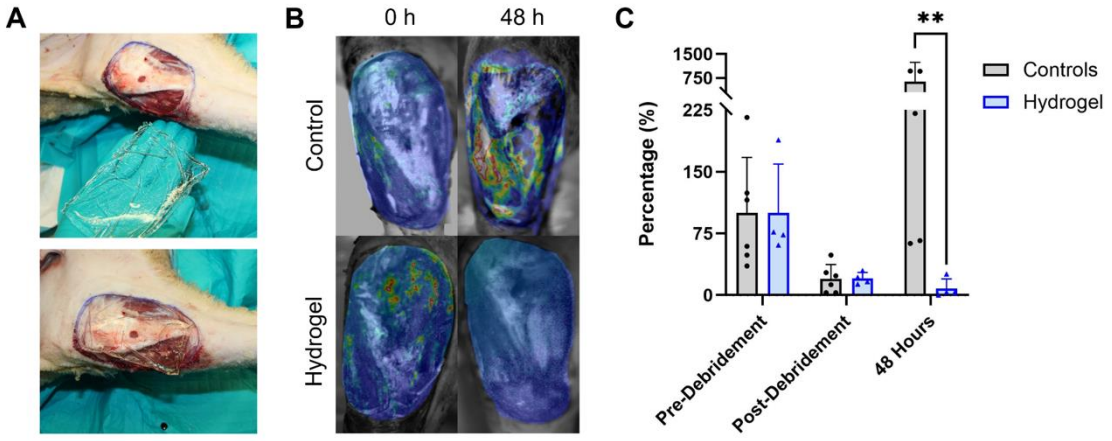




In vivo antibiotic efficacy in murine models

In vitro antibiotic efficacy

	MIC (CLSI/Lit.)	Tobramycin Hydrogel	Vancomycin Hydrogel	Tobramycin + Vancomycin Hydrogel
S. Aureus	≤ 2 μg/mL	–	4.60 ± 1.50 μg/mL	3.60 ± 0.30 μg/mL
E. Coli	≤1 μg/mL	7.48 ± 1.70 μg/mL	–	5.37 ± 0.50 μg/mL



In vivo antibiotic efficacy in ovine models





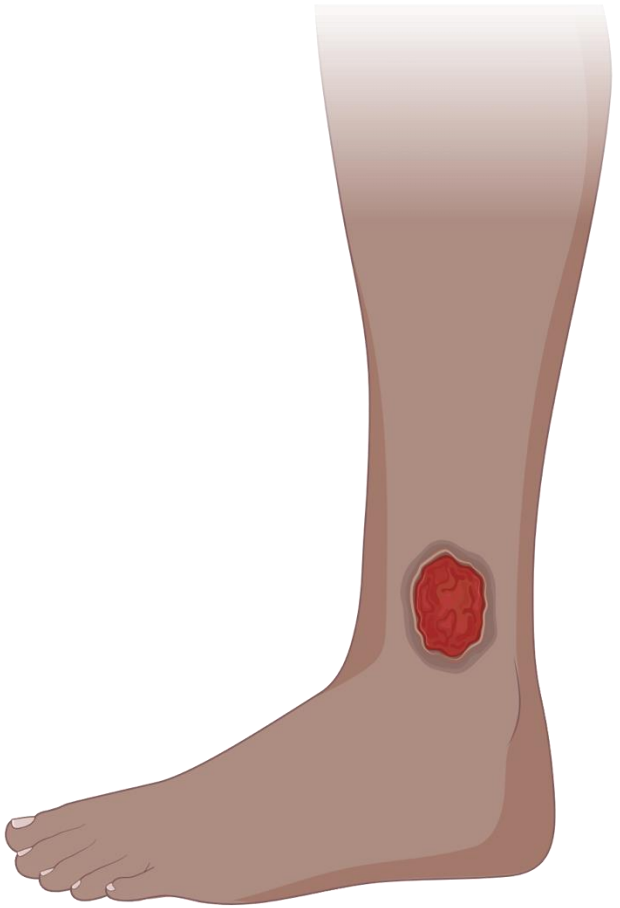


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Multifunctional chitosan hydrogels for SCLUs

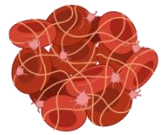
OVERVIEW OF SCLU

- Different from most chronic wounds
  - Stems from vaso-occlusion
- High susceptibility to microbial infections
- Occurs in the lower extremities





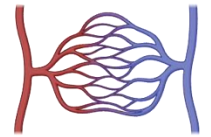
Endothelial dysfunction & venous insufficiency



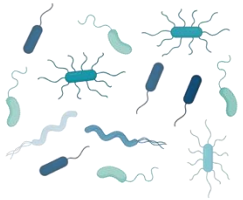
Thrombosis



Altered rheology, hemolysis



Decreased oxygen delivery to skin



Infection (*S. aureus, Pseudomonas spp.,*

Pathogenesis of skin ulceration and delayed healing

Bonkemeyer Millan *et al., 2019. Am. Fam. Physician*
Kato *et al.*, 2018. *Nat. Rev. Dis. Primer*
Laudin *et al., J. Bone Jt. Surg.* Images created with BioRender

Why chitosan hydrogels for SCLUs?

CURRENT SCLU THERAPY

- Foams and hydrocolloids
- Iodine and silver dressings
- Debridement
- Drugs e.g., topical sodium nitrite

CRITICAL NEEDS

- All-in-one therapy
- Addressing the underlying SCLU drivers
- Cost and accessibility

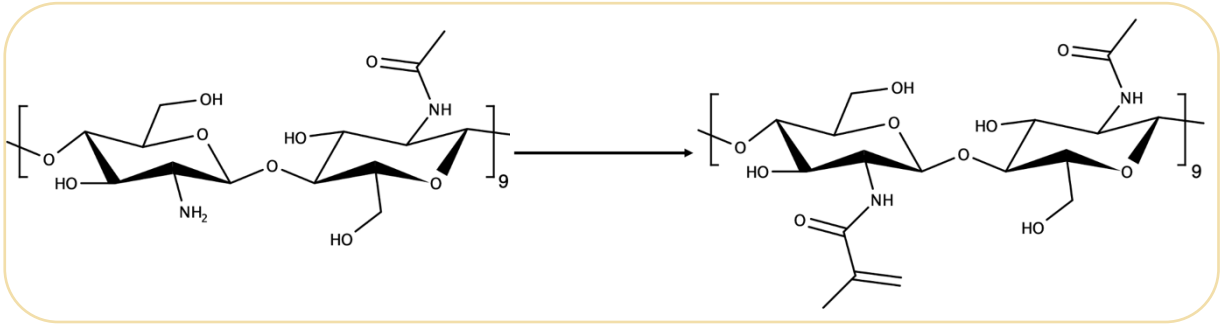
OUR DESIGN

- ✓ Chitosan
  - GRAS
  - Natural antimicrobial agent
  - Cheap and readily available
- ✓ PEGDA
  - Biocompatible and FDA approved
- ✓ UV-crosslinkable system with Irgacure 2959
  - Cheap, biocompatible
- ✓ Release of a vasodilator
- ✓ Designed for real world-use in low resource settings

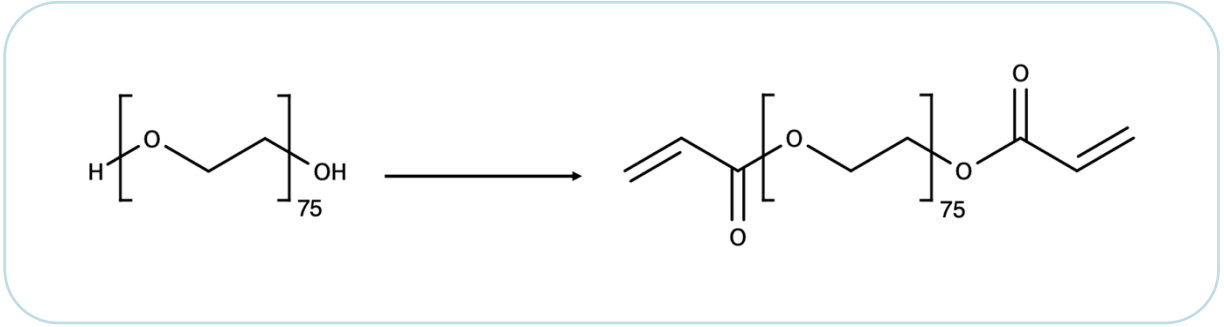
Millan *et al., 2019. Am. Fam. Physician .*
Minnit, C.P *et al.,* 2014. *Lancet Hematol.*

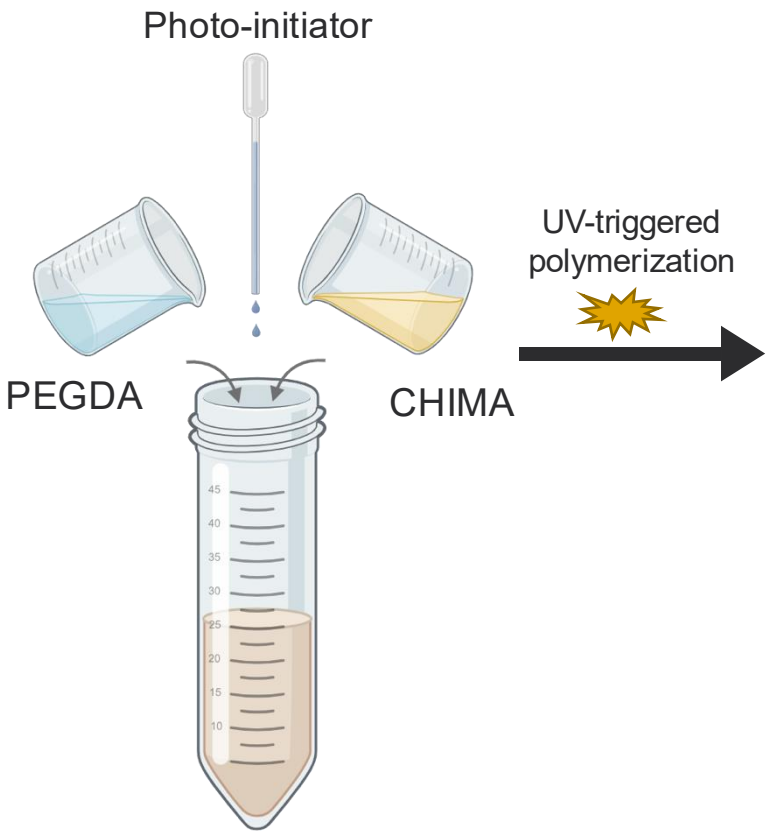
Hydrogel design and synthesis

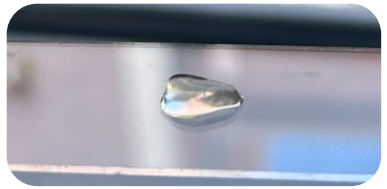
Chitosan methacrylate (CHIMA)



Polyethylene glycol diacrylate (PEGDA)

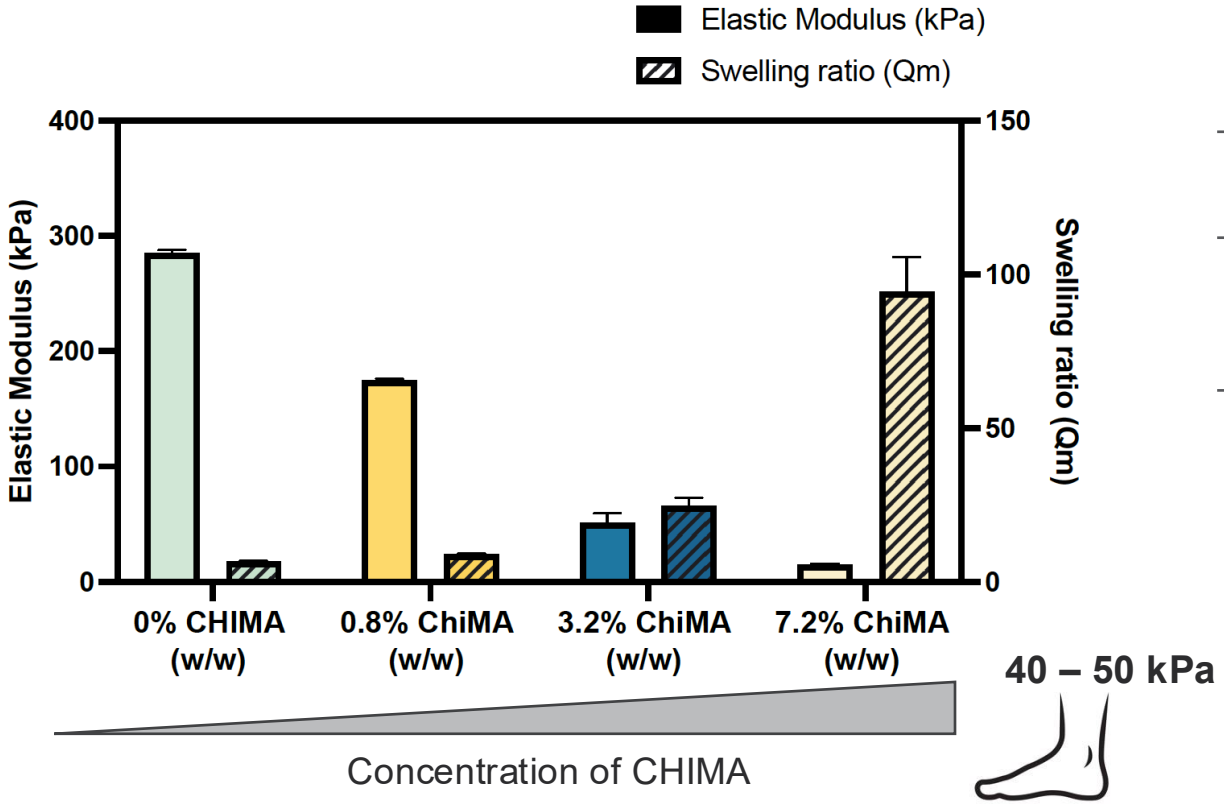






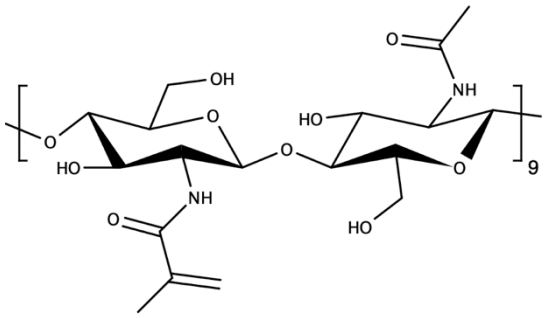
Chitosan hydrogels...

AS A TUNABLE SKIN ANALOG



Experimental group	Mesh size (Å)	Relative swelling capacity (%)
0.8% CHIMA (w/w)	66.0 ± 1.5	104.0
3.2% CHIMA (w/w)	165 ± 13	388.0
7.2% CHIMA (w/w)	394 ± 41	1844

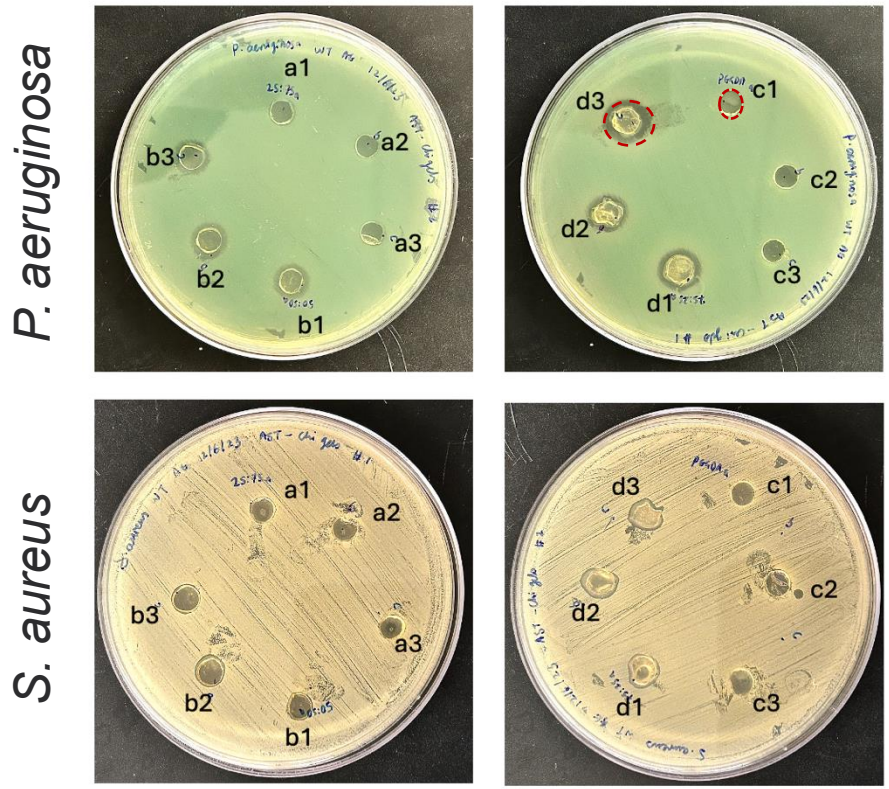




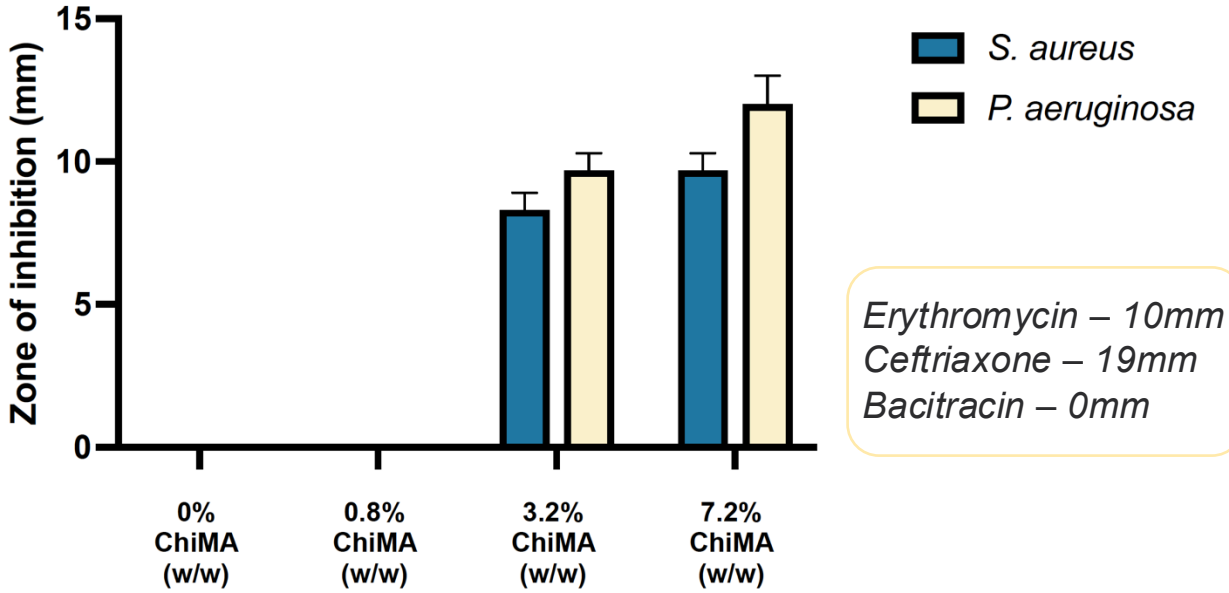
Elastic modulus and swelling ratio trends of CHIMA hydrogels

Chitosan hydrogels...

AS A NATURAL ANTIMICROBIAL AGENT



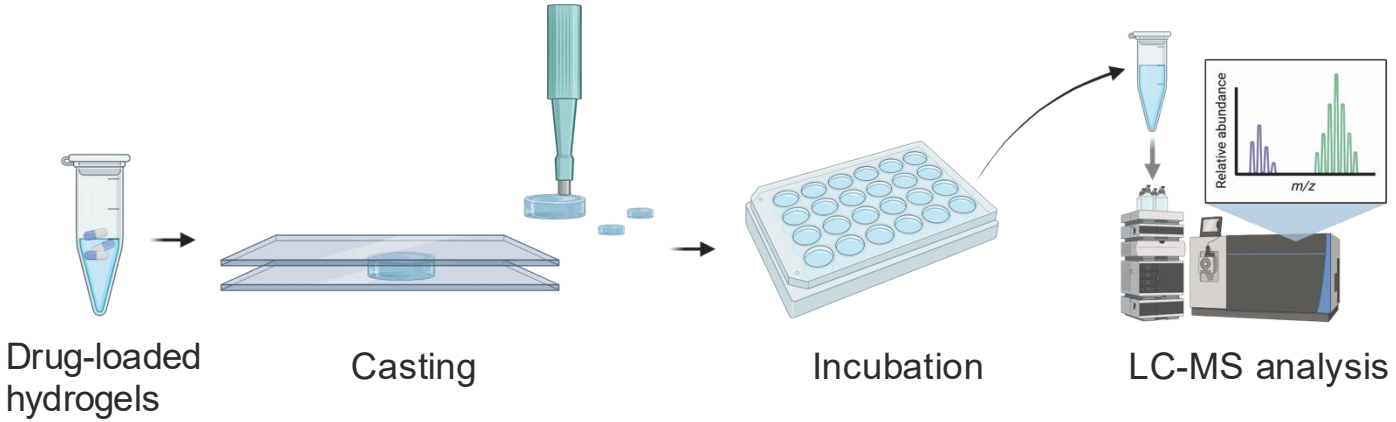
Images of bacterial culture plates after ZOI experiments
a = 0.8% CHIMA gel, b = 3.2% CHIMA gel, c = 0% CHIMA/ control, d = 7.2% CHIMA gel

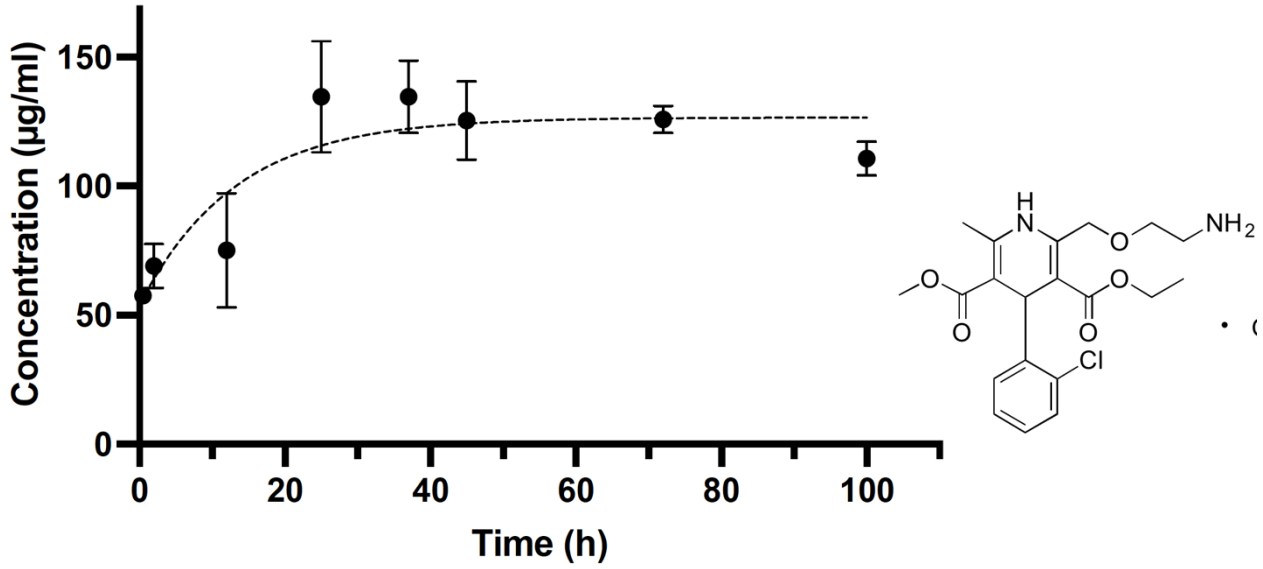


Quantification of the ZOI studies

Chitosan hydrogels...

AS A RELEASE VEHICLE



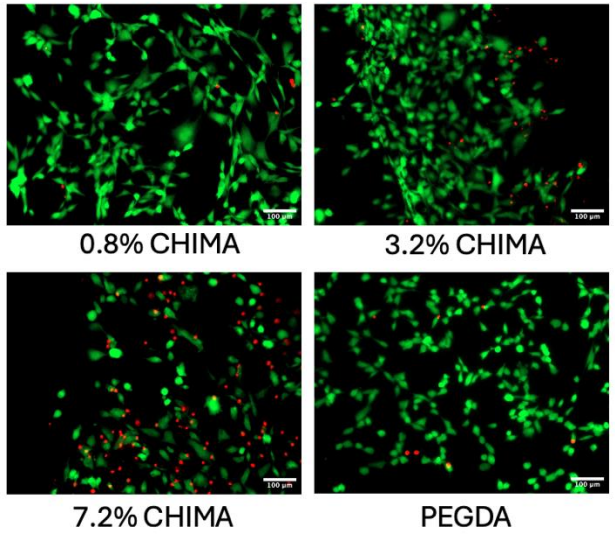


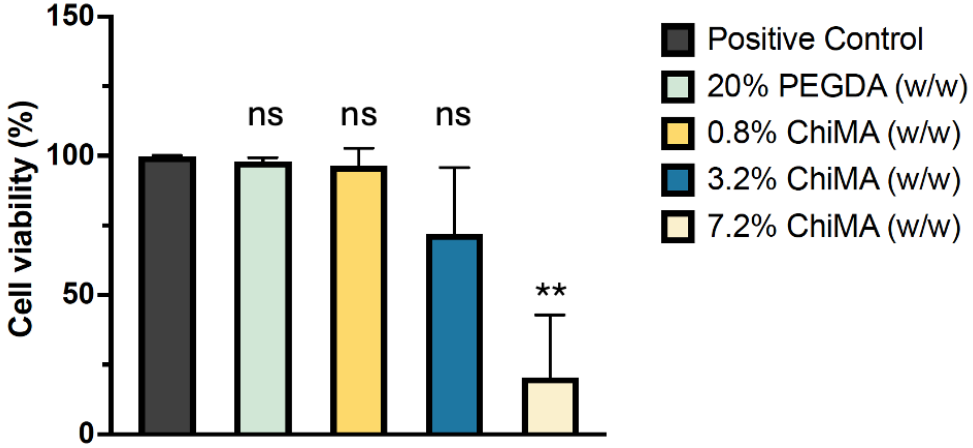
Release of a vasodilator (amlodipine)

Chitosan hydrogels...

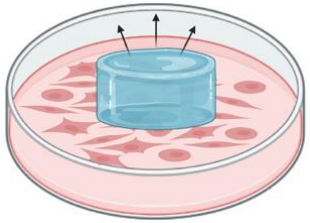
AS A CYTOCOMPATIBLE MATRIX

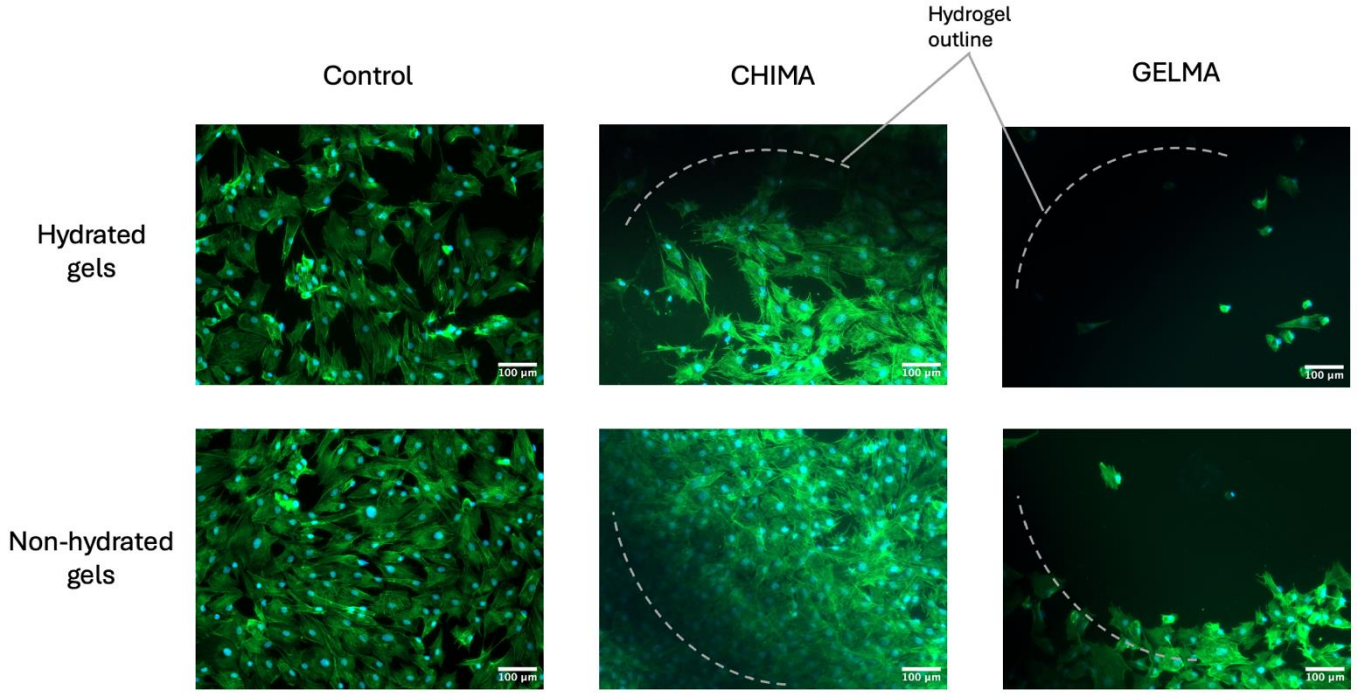
NIH mouse 3T3 fibroblasts





Hydrogel-cell compatibility





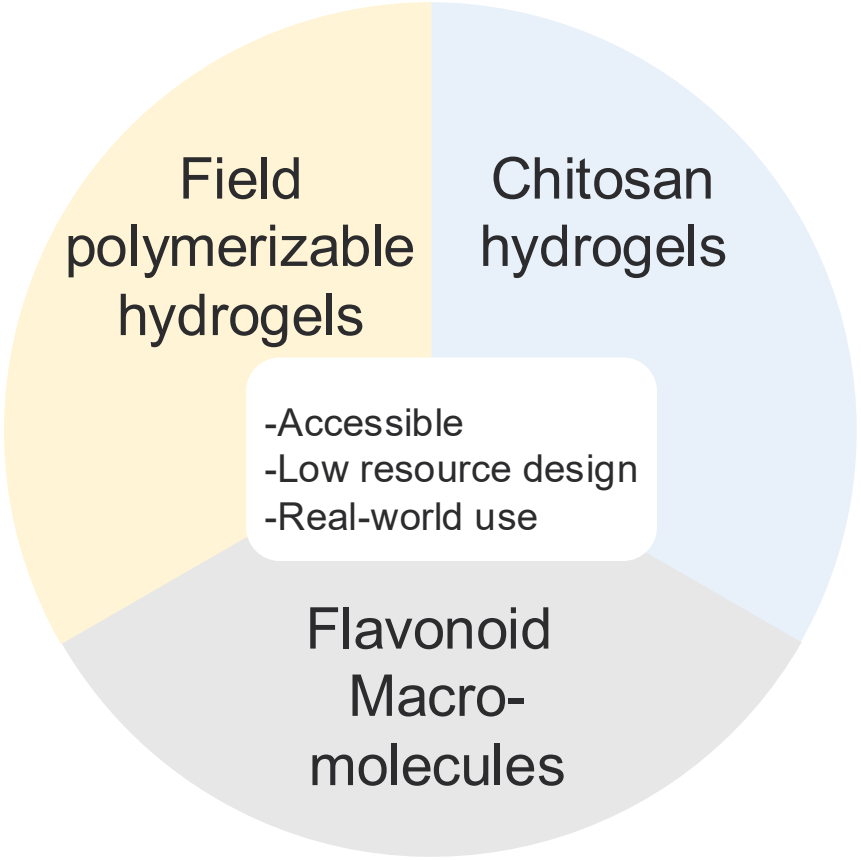
NIH mouse 3T3 fibroblasts

Hydrogel-cell adhesion testing

Research Themes, Broader Impact and Future Perspectives



Pushing the boundaries: biodegradation



Other venous ulcers

Q&A









